Large Language Models for Bayesian Network Parameterization in Software Engineering: Direct Elicitation versus Ranked Nodes

Anonymous Author(s)

Abstract

Software engineering (SE) decision makers need models they can understand and use when data are scarce. Bayesian networks (BNs) make probabilistic relationships explicit and incorporate expert knowledge, but eliciting their conditional probability tables (CPTs) is costly. The Ranked Nodes Method (RNM) reduces the required input by generating ordinal CPTs from partial judgments. Large language models (LLMs) offer another source of judgments: should they fill complete tables or supply fewer inputs to RNM? In an exploratory study, we compared these approaches on every row of 61 ordinal conditional nodes from seven SE BNs. Structured prompting and qualitative retrieval-augmented generation (RAG) formed additional factors in eight conditions, each repeated five times using GPT-6 Astra at medium effort. With equal node weights, mean total variation was 0.2317 for direct elicitation and 0.2404 for RNM, a 3.76% increase. Because nodes were unevenly distributed across five source studies, we also weighted each study equally; direct elicitation still had lower error. Both contrasts passed Holm-adjusted paired t-tests, but not exact sign tests. RNM improved 31 nodes, yet larger deteriorations elsewhere outweighed those gains. Structured prompting modestly helped direct elicitation under node weighting; qualitative RAG demonstrated no benefit. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, but only 0.61% fewer total tokens. The findings favor direct elicitation as the average numerical baseline in this sample and show why reduced elicitation alone cannot establish a better, cheaper to complete CPT.

CCS Concepts

• **Computing methodologies** → **Bayesian networks**; • **Software and its engineering** → *Software creation and management*.

Keywords

Bayesian networks, large language models, probability elicitation, ranked nodes, software engineering

1 Introduction

Supporting software engineering (SE) decisions faces two main challenges: explaining the basis for a recommendation and obtaining enough relevant data to build a model experts can rely on. Examples include release readiness, quality assessment, and process improvement, where decision makers must judge whether a recommendation's assumptions apply. A review of 108 studies documents the demand for explainability across 23 SE tasks [3]. Training data for machine learning and deep learning may also be scarce, particularly for managerial SE decisions [23]. Explicit assumptions and domain knowledge are therefore valuable.

Bayesian networks (BNs) offer such an approach. A BN makes relationships and uncertainty explicit in a graph and probabilities, incorporating domain knowledge when data are limited [10, 11]. BNs have supported many SE problems, including early defect prediction [10], effort estimation [17], and diagnosis of development-process problems [21]. A systematic mapping of 106 studies identifies software management and quality as prominent application areas [24]. Building a credible BN nevertheless requires substantial knowledge-engineering effort.

Constructing conditional probability tables (CPTs) is a major part of that effort. Each row describes a child's probabilities for one combination of its parents' states; adding parents rapidly increases the number of judgments required [11]. Semi-automatic methods reduce this burden by reconstructing a table from fewer expert judgments or parameters. These include the Ranked Nodes Method (RNM) [11], the Weighted Sum Algorithm (WSA) [20], Analytic Hierarchy Process (AHP)-based pairwise comparisons [5], and interpolation methods [1, 2]. Their assumptions can, however, limit fidelity to the intended probabilities [2].

Large language models (LLMs) are now being explored for CPT construction. Nafar et al. elicited numerical BN parameters directly [18]; Gottal and Matthes reconstructed CPTs from statistical summaries extracted from scientific literature [12]; and Cypko et al. used retrieval-augmented generation (RAG) to support expert BN modeling [6]. Should an LLM specify every row, or supply fewer judgments for a semi-automatic method to complete? Any benefit depends on both the judgments and the reconstruction.

We investigated this question using RNM, which applies to variables with ordered states such as Low, Medium, and High [11, 15]. Instead of requesting the entire CPT, we asked the LLM for distributions at selected parent-state combinations, called *anchors*, and for a rule describing how the parents combine. A numerical fitting procedure then adjusted that rule's weights and uncertainty to match the elicited distributions and generated the complete CPT. We compared this approach with direct complete-table elicitation on 61 ordinal conditional nodes from seven SE BNs.

We also examined support for elicitation. Structured prompts made conditioning explicit and asked for probability and consistency checks; qualitative RAG supplied short passages explaining relevant domain relationships. Crossing the two elicitation methods, two prompt designs, and presence or absence of RAG produced eight conditions. We repeated each five times using GPT-6 Astra at medium effort. This study addresses three research questions:

- **RQ1—Reference agreement:** How does LLM-assisted RNM compare with direct elicitation in complete-CPT agreement across the full SE sample?
- **RQ2—Elicitation support:** How do structured prompting and qualitative RAG affect each method's agreement across complete tables?
- **RQ3—Accuracy and effort:** How does RNM trade reference agreement against requested judgments and observed computational effort?

SAC '27, Gwangju, South Korea
2027.

All three RQs include all 61 nodes and every CPT row, including anchors. The seven BNs cover Scrum process assessment, Chrome code review, ERP data quality, software maintenance logs, and software strategic indicators. We give nodes equal weight for the main comparison. Because BNs contribute unequal numbers of nodes, we also weight their five source studies equally and report each BN separately. This evaluates the complete tables available for use rather than restricting attention to configurations omitted from RNM prompts.

Direct elicitation has lower mean complete-table TV: 0.2317 versus 0.2404 for RNM. The direction persists under equal-family weighting and all four prompt/evidence combinations, although individual nodes and BNs differ. Structured prompting offers limited benefit for direct elicitation; qualitative RAG does not demonstrate improvement. RNM requests 40.1% fewer distributions but uses only 0.61% fewer total tokens. Our contribution compares the complete construction pipelines and their computational effort, with fitting diagnostics explaining possible sources of variation.

Section 2 positions the study in related work, Section 3 describes the experiment, and Section 4 reports the results. Section 5 discusses the RQs and implications for SE, Section 6 examines validity threats, and Section 7 concludes the paper.

2 Related Work

Table 1 positions our comparison by what the LLM supplies and how its output is evaluated. Numerical CPT agreement, modeling support, and downstream predictive performance address different questions.

2.1 Numerical elicitation versus modeling support

Nafar et al.'s broad benchmark is the closest numerical comparison [18]. Requesting one child-state probability at a time lowered median BN KL divergence for seven of eight models, at the cost of more queries. Their FullDist baseline requests one conditional row per query; ours requests the whole CPT. Their granularity result does not determine whether omitting entire configurations is beneficial. Gottal and Matthes instead reconstruct from numerical summaries [12], which impose constraints absent from our qualitative RAG packets. Rahul and Chowdhury's persona-panel preprint reports effort and internal consistency, including refitting self-generated data [22]; this cannot establish independent reference agreement.

BIRD combines neural and symbolic reasoning [7] by combining LLM judgments, constrained fitting, and BN inference [9]. Its downstream evaluation differs from exact CPT recovery. Clinical RAG assists modelers [6], while AutoElicit evaluates prediction after updating elicited parameter priors [4]. Our endpoint is numerical agreement, not explanation quality or practitioner benefit.

2.2 Partial elicitation and our contribution

RNM constructs ordinal CPTs from weighted parent functions and truncated-normal distributions [11, 15]; InterBeta offers another partial-elicitation approach [1]. Blomaard et al. compare RNM, functional interpolation, and InterBeta using known CPTs to optimize parameters [2]. A recent preprint compares human-elicited WSA and RNM in software decisions [20]. Our experimental fits instead use LLM anchors alone; reference-informed fits remain separate diagnostics.

The unresolved comparison is whether sparse LLM judgments plus reconstruction retain agreement while reducing effort. We compare complete CPTs across the full sample and use same-anchor diagnostics to examine changes introduced by fitting. This distinguishes elicitation from fitting without assuming that ordinal eligibility guarantees RNM compatibility or that fewer distributions imply proportional token savings.

3 Method

We compared direct CPT elicitation with LLM-assisted RNM reconstruction in a repeated $2 \times 2 \times 2$ design: method (direct/RNM), prompt (baseline/structured), and evidence (no RAG/qualitative RAG). The directional expectations were better reference agreement with RNM (RQ1), improved agreement from elicitation support (RQ2), and fewer requested distributions and output tokens with RNM (RQ3). Graphs, definitions, state mappings, prompts, and numerical procedures were fixed before collection.

3.1 Sample and references

We selected 61 conditional targets from seven BNs associated with published SE studies with complete numerical references and interpretable, ordinal child and parent states (Table 2). Binary variables qualified only when their meanings supported an order; nominal families and Boolean gate chains were excluded. This availability-based sample covers Scrum practices [19], Chrome code review and defects [14], ERP data quality for advanced planning and scheduling (APS) [13], software maintenance logs [8], and software strategic indicators (SESSI) [16]. The two Chrome versions share one publication family, as do the two log BNs.

The Scrum reference implements a published framework relating team and process factors to sprint and backlog quality [19]. Its archived XML has 25 conditional nodes, 17 roots, and three states, whereas the publication describes 44 nodes and five states. We reconstructed 249 conditional rows from the XML's functions, weights, variances, and expressions. Scrum is therefore an RNM-derived reference; the other BNs provide exported CPTs, with SESSI using the original elicited table in Appendix A3. We checked dimensions, normalization, and indexing. The largest retained CPT has 27 rows. The supplement specifies reconstruction details.

All three RQs use the full 61-node sample. Agreement is scored on complete CPTs, including anchors and configurations not elicited in RNM prompts. This evaluates the table available for BN use, rather than only its completion beyond elicited inputs. Anchors remain legitimate evaluation cases: RNM fits LLM judgments, not reference probabilities. Nodes with one parent or with all rows elicited are retained because fitting can still change their probabilities. References describe particular modeling contexts, not universal company probabilities. The complete-table endpoint, weighting, and tests were adopted after inspecting earlier results; this analysis is exploratory.

3.2 What the LLM received and supplied

All conditions shared an expert-role description, setting, variable definitions, state labels and order, and assumed parent influence

Table 1: LLM-assisted BN construction compared by inputs, evaluation, and scope. Entries distinguish numerical CPT agreement from modeling support and downstream performance; reported errors are not directly comparable. KL denotes Kullback–Leibler divergence.

Study	Construction and supplied evidence	Evaluation material and endpoint	Scope relative to our comparison
Nafar et al. [18] TMLR, 2026	LLM internal knowledge; one conditional row or one child-state probability per query. A separate stage combines estimates with observations.	80 public BNs, eight LLMs: BN/CPT KL. Separate classification evaluation on nine datasets with three LLMs.	Changes granularity within a row; our RNM approach requests selected rows and reconstructs the rest.
Gottal and Matthes [12] Findings ACL, 2026	Reconstruction algorithms and direct LLM CPT output from numerical statistical summaries; separate extraction from publications.	Synthetic BNs and publication/raw-data cases; CPT distances, inference and decision utility, and runtime.	Quantitative summaries constrain construction; our RAG packets supply qualitative context.
Rahul and Chowdhury [22] Preprint, 2026	Persona-conditioned LLM panels provide distributions for parent configurations; trimmed-mean aggregation.	One healthcare-consultation BN; plausibility and internal consistency checks; calls, token estimates, and runtime.	Demonstrates construction feasibility and effort; lacks an independent numerical CPT benchmark.
Feng et al. (BIRD) [9] ICLR, 2025	LLM-generated factors and coarse likelihood judgments feed constrained parameter fitting and BN inference.	Human judgments of relative support and downstream task accuracy.	Establishes a precedent for LLM judgments plus fitting; does not test numerical reference-CPT recovery.
Cypko et al. [6] IJCARS, 2026	Clinical RAG supports expert BN modeling using guidelines and textbooks.	Synthetic retrieval evaluation; modeling, usability, and workload study with four clinicians.	Tests modeling support rather than complete numerical CPT agreement.
This study	Complete CPT elicitation versus LLM anchors plus RNM fitting; baseline/structured prompts, with/without qualitative RAG.	61 ordinal conditional nodes from seven SE BNs; complete-table distances, five replications, tokens, and fitting time.	Compares final CPTs on all rows of all nodes; checks weighting sensitivity and uses anchor fitting as a diagnostic.

Table 2: All 61 evaluated targets and requested distributions per prompt/evidence combination and replication. Every CPT row contributes to agreement scores; anchors are RNM fitting inputs.

Network	Targets	CPT rows	Anchors
Scrum	25	249	123
Chrome 40	15	78	57
Chrome 42	12	63	46
APS	3	8	8
SESSI	1	3	3
Software logs 3	4	16	12
Software logs 4	1	4	3
Total	61	421	252

directions: whether higher parent states should tend to increase or decrease the child. Each distribution conditioned on *all* listed parent states and averaged over unspecified circumstances. The persona supplied a professional perspective without claiming company observations.

Baseline prompts requested normalized distributions in JSON. Structured prompts additionally made conditioning explicit, framed probabilities as expected counts among 100 hypothetical comparable cases, distinguished evidence from assumptions, and requested consistency checks. Fractional counts were allowed; these cases aided judgment rather than supplying observations. A brief evidence note was requested, without private reasoning traces.

Direct elicitation requested every CPT row in one call. RNM requested predetermined anchor distributions and a combination rule: weighted mean, weighted minimum, weighted maximum, or minimum–maximum mixture. These express different degrees of compensation between stronger and weaker parents. A numerical fitter then estimated weights and variance from the LLM's answers. With one parent, its direction-aligned value supplied the distribution mean directly. Both methods received the same influence assumptions; only RNM enforced them mathematically. Of 92 directions, nine followed from definitions, 43 had qualitative support, and 40 were labeled uncertain hypotheses.

RAG supplied fixed cards paraphrasing relevant domain explanations. One complete card for Scrum Cohesion stated:

> **Shared goals.** Cohesive teamwork includes commitment to shared tasks and team goals. Agile collaboration involves joint responsibility, rather than treating every task as an isolated individual obligation.

This adds context, not frequencies for Low, Medium, or High cohesion. Retrieval matched cards to model context and target/parent tags, selecting up to three; packets contained 24–103 words (median 72). Each node's packet was reused across paired conditions and replications. Cards excluded reference probabilities, numerical parameters, and identifying artifact links. Both arms already received definitions, so RAG tested additional qualitative context,

not live retrieval or company measurements. Input budgets were not equalized.

3.3 From anchors to complete tables

RNM mapped ordered states to equal intervals in $[0, 1]$, reversed negative parent influences, and averaged truncated-normal probabilities over five integration points per parent interval [11, 15]. Anchors comprised all-low, all-high, all-middle, one-high/rest-low, and one-low/rest-high configurations, removing duplicates without consulting the reference. Two ternary parents, for example, yield nine rows and five anchors: $(0, 0)$, $(2, 2)$, $(1, 1)$, $(2, 0)$, and $(0, 2)$. RNM generated a nine-row table from those five distributions. RQ1 compared all nine reconstructed rows with all nine directly elicited rows against the same reference.

For the LLM-selected rule, deterministic multistart search minimized mean anchor TV between elicited and fitted distributions, without reference access. Valid poor fits were retained. The supplement provides equations, parameter bounds, integration, and search settings. Agreement does not establish recovery of unique parameters.

3.4 Execution and reference isolation

Five replications were collected on 15–16 October 2026 using Codex CLI 0.162.1, requesting `gpt-6-astra` at `medium` effort. Each target–condition–replication used a fresh session; requests were shuffled within replications. Temperature and generation seed were unspecified. Experimental sessions could not access references, full papers, evaluator code, or authoring conversations; tools, browsing, connectors, and memory were disabled.

Outputs required correct rows/states, finite probabilities in $[0, 1]$, and sums within 10^{-6} of one. One fresh-session format repair was allowed without reference feedback. All 2,440 trials produced valid outputs after four repairs (two per method); accounting includes all 2,444 attempts, with no fallback distributions.

3.5 Outcomes and analysis

For reference row p and prediction q, lower total variation means closer agreement:

$$\mathrm{TV}(p, q) = \frac{1}{2} \sum_{s} |p_s - q_s| \in [0, 1]. \tag{1}$$

For each node, we average TV over all its rows. The primary full-sample score gives each of the 61 nodes equal weight; it therefore prevents larger CPTs from dominating solely through their row counts. RQ1 compares RNM minus direct after equally averaging the four prompt/evidence combinations within each replication. RQ2 contrasts structured minus baseline and RAG minus no RAG within each method, equally averaging the other factor. All eight condition means use the same complete-table endpoint.

The sample is unevenly represented: Scrum and the two Chrome BNs supply 52 of 61 nodes, so they dominate the node-weighted mean. To check this influence, we group BNs from the same publication into a family and give the five families equal weight, averaging nodes within BNs and BNs within families. This retains all nodes while reducing the influence of studies contributing more nodes. We report all seven BNs descriptively to expose variation behind either aggregate. Jensen–Shannon divergence, ordinal-CDF distance, and smoothed CPT KL use the same complete-table weights in supplementary results.

RQ3 counts requested distributions before repairs and client-reported tokens over all attempts for the full sample. Total tokens equal input plus output; cached-input and reasoning-output counters are subsets, not additional tokens. Paired contrasts use replication totals. Latency and offline fitting time are descriptive; subscription tokens are not converted into monetary savings.

The five paired replication blocks are the inferential units, not correlated nodes or rows. Two-sided paired t-tests ($df = 4$) and pointwise 95% intervals assume approximately normal, independent, stationary block differences. Five blocks provide little evidence for these assumptions. Two-sided exact sign tests assess direction/median sensitivity under independence; five concordant nonzero differences still yield $p = 0.0625$. Holm adjustments cover two RQ1 weighting contrasts, eight RQ2 contrasts (two methods, two factors, two weighting rules), and two RQ3 token contrasts. These are separate exploratory families, not study-wide confirmatory control; intervals are not simultaneous. No prespecified practical-effect or equivalence margin supports claims of usefulness or equivalence.

3.6 Offline diagnostics

Supplementary post-hoc diagnostics compare raw/fitted anchors, uniform predictions, alternative families, direct-anchor fits, stronger searches, and wider bounds. Reference-informed fits assess best-found approximation, not held-out accuracy or error floors. Non-anchor, Chrome-version, and node-omission diagnostics concern a 25-node compression subset. Their rows and weights are specified separately; they neither define the main endpoint nor replace its predictions.

4 Results

RNM requests fewer distributions but increases mean complete-table disagreement. All main results include every row of all 61 nodes; lower TV is better, and negative RNM-minus-direct differences favor RNM.

4.1 RQ1: Which method produces closer complete CPTs?

Direct elicitation has lower mean error across the sample. Mean TV is 0.2317 for direct elicitation and 0.2404 for RNM, an absolute increase of 0.0087 or 3.76% with RNM. The paired difference is positive in all five replications, ranging from 0.0072 to 0.0094 (Figure 1b). Its pointwise 95% t interval is $[0.0076, 0.0098]$, with Holm-adjusted $p = 0.000048$. The corresponding exact sign test does not reject ($p = 0.125$ after adjustment). Observed differences are similar across the five runs, but these blocks provide limited distribution-free evidence. Neither test establishes whether the difference matters for a particular SE decision.

Because Scrum and Chrome supply 52 of 61 nodes, we checked whether their numerical dominance explains the result. Giving each of the five source studies equal weight still favors direct elicitation: TV is 0.1937 for direct elicitation and 0.2189 for RNM, a difference of 0.0252 or 12.99%. Its interval is $[0.0211, 0.0292]$, adjusted t $p = 0.000065$, and adjusted sign $p = 0.125$. This is the same 61-node

Table 3: Complete-table TV by BN, averaged over prompt/evidence conditions and replications. Δ is RNM minus direct. Every node is included; BN rows are descriptive.

BN	Nodes	Direct	RNM	Δ
Scrum	25	0.2605	0.2334	−0.0272
Chrome 40	15	0.2069	0.2220	+0.0151
Chrome 42	12	0.1988	0.2335	+0.0347
APS	3	0.0964	0.1155	+0.0191
SESSI	1	0.1160	0.1143	−0.0017
Software logs 3	4	0.3802	0.4770	+0.0967
Software logs 4	1	0.2052	0.3299	+0.1247
All nodes	61	0.2317	0.2404	+0.0087

sample under different weights. The direction therefore persists when studies contributing more nodes receive less influence. Secondary complete-table distances also favor direct elicitation under both weighting rules (supplement).

The aggregate reflects unequal gains and losses. Table 3 and Figure 1a show every BN. RNM improves mean agreement for Scrum and slightly for SESSI, while both Chrome versions, APS, and both software-log BNs worsen. At node level, RNM is closer for 31 of 61 nodes and direct elicitation for 30. The overall mean favors direct elicitation because the magnitudes of deterioration outweigh the improvements; it is not a count of node-level wins. These descriptive comparisons reveal variation rather than establishing seven separate population effects.

Fitting can improve one relationship and weaken another. Consider a Scrum team with low self-organization but medium collaboration. Its Cohesion node represents working synchronously toward shared goals. For this configuration, the reference assigns 76.6% probability to Low cohesion. Direct elicitation assigns 58.0%, while RNM reconstructs 67.6%, moving closer to the reference (Figure 2a). Mean row TV falls from 0.1862 to 0.0901. Here, a relationship fitted to other elicited configurations improves a row of the final CPT.

Now consider a Chrome component with three or four contributors and no minor contributors. Contributors are classified by their share of changes. With no minor contributors, all must be major, so the Major Contributors node belongs entirely in the three-to-four category. Both the reference and all direct responses assign probability one to that category. RNM assigns 84.4%, spreading the remaining 15.6% to neighboring categories (Figure 2b). Smoothing through the fitted family weakens a count constraint preserved by direct elicitation. These deliberately contrasting examples explain possible changes, not their prevalence across all seven BNs.

Supplementary diagnostics compare raw and fitted judgments on identical anchors in the 25-node compression subset. Fitting changes Scrum anchor TV little (0.2790 to 0.2762) but raises Chrome anchor TV (0.2491 to 0.2994). Alternative families and searches, reference-informed approximation, and node-omission checks help examine that contrast, with their restricted scope stated separately. They do not establish a single cause of the full-sample result. **RQ1 does not support an overall RNM advantage: direct elicitation has lower mean complete-table TV under both weighting rules, despite improvements in some nodes and BNs.**

Table 4: All eight conditions: complete-table TV across all 61 nodes, with equal node weights and five replications. Each row pairs the methods under the same prompt and evidence.

Prompt	Evidence	Direct	RNM	Δ
Baseline	None	0.2327	0.2396	+0.0070
Baseline	RAG	0.2349	0.2419	+0.0070
Structured	None	0.2282	0.2389	+0.0107
Structured	RAG	0.2311	0.2413	+0.0102

4.2 RQ2: Do richer prompts or qualitative evidence help?

Direct elicitation has lower mean TV in every matched prompt/evidence combination (Table 4). Structured prompting lowers the direct method's node-weighted TV by 0.00413, with interval [−0.00595, −0.00232] and Holm-adjusted t $p = 0.0255$. This is the only RQ2 contrast that rejects after the eight-test correction. Under equal-family weighting, the direct prompt effect is −0.00508, but its interval crosses zero and adjusted $p = 0.4953$. Thus, the point estimates agree in direction while uncertainty differs with weighting. No RQ2 exact sign test rejects after adjustment.

For RNM, the node-weighted structured-prompt effect is −0.00067. Adding RAG changes TV by +0.00252 for direct elicitation and +0.00230 for RNM; none of these adjusted tests rejects. The Cohesion packet, for example, describes shared goals without stating how often cohesion is Low given Low self-organization. It adds context to definitions rather than numerical constraints. **RQ2 provides limited support for structured direct elicitation, but no demonstrated benefit from the tested qualitative RAG.** This does not establish equivalence or the ineffectiveness of local quantitative evidence. Full contrasts under both weights appear in the supplement.

4.3 RQ3: Do fewer judgments translate into fewer tokens?

RNM requests 5,040 distributions across the experiment, compared with 8,420 for direct elicitation: 40.1% fewer. Both methods still make one initial request per node and condition. Reducing the returned table therefore does not eliminate the shared instructions, definitions, and evidence sent with each request.

Output tokens fall by 12.12%, but input-plus-output tokens fall by only 0.61% (Table 5). Inputs dominate: approximately 14.36 million for direct elicitation and 14.31 million for RNM. Across five paired replication totals, output savings have adjusted t $p = 0.000073$, whereas total-token savings do not reject ($p = 0.0809$). Both adjusted exact sign tests give $p = 0.125$. The supplement reports block differences and intervals. Fewer requested probabilities therefore do not yield proportional computational savings.

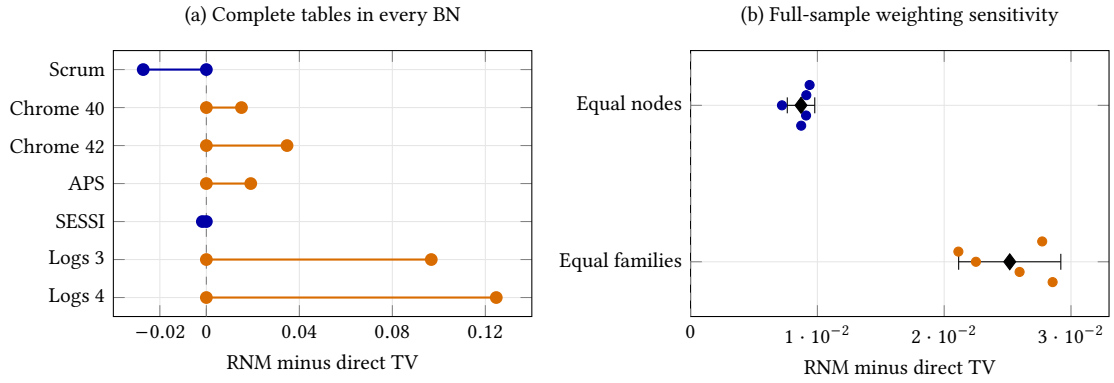


Figure 1: Complete-CPT agreement across the full sample. Negative differences favor RNM. (a) Each BN's mean averages all rows within nodes, then nodes, four prompt/evidence combinations, and five replications equally; points are descriptive. (b) Both weighting rules retain all 61 nodes. Circles show the five block differences; diamonds and whiskers show means and pointwise 95% t intervals. Equal-family weighting first averages nodes within BNs and BNs within each of five publication families.

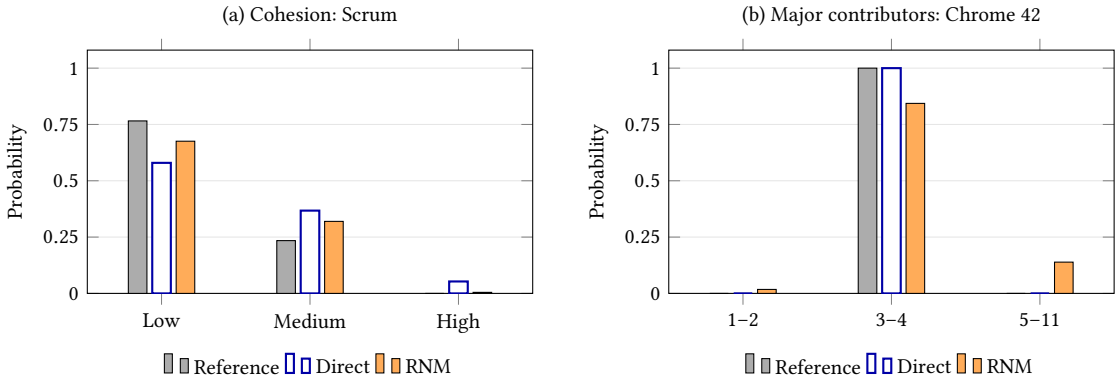


Figure 2: Two CPT rows unqueried by RNM. (a) Cohesion given Low Self-Organizing and Medium Collaboration. (b) Major contributors given three or four total contributors and no minor contributors. Bars average 20 predictions per method. Examples were selected for opposing effects in the earlier non-anchor comparison; they illustrate mechanisms and do not represent the full sample.

Table 5: RQ3: tokens across all 61 nodes and five replications, including repairs. Holm-adjusted paired-t tests compare five method-paired block totals across two counters; neither exact sign test rejects.

Counter	Direct	RNM	Reduction	Holm t p
Output	341,153	299,798	12.12%	0.000073
Input + output	14,697,090	14,607,517	0.61%	0.080921

Summed request latency is 3.82% lower for RNM; the original retained fits sum to 182.3 seconds of recorded fitting time, excluding subsequent offline diagnostics. These are computational descriptions, not controlled wall-clock or monetary savings. **RQ3 supports reduced elicitation and output size, with limited evidence of total-token savings.** Those savings accompany higher mean complete-table error across the sample, so they do not establish a superior accuracy–effort tradeoff.

5 Discussion

The full-sample comparison exposes an accuracy–effort tradeoff: RNM requests fewer judgments and produces shorter answers, yet its final CPTs have higher mean reference error. Network-level results show where that summary applies and where it does not. We interpret the three RQs against prior work, then consider SE research and practice.

5.1 Research questions in relation to the literature

RQ1: Reference agreement. RNM's partial-elicitation rationale [11, 15] does not ensure that fitting LLM judgments improves final CPTs. Direct elicitation has lower mean error under both full-sample weighting rules, but RNM is closer on a small majority of nodes. The magnitude and distribution of errors therefore matter more than a count of wins. Equal-family weighting increases RNM's disadvantage because the smaller software-log BNs have large deteriorations, while the 25-node Scrum BN improves. Neither weighting estimates a universal effect across SE organizations.

Compatibility between the reconstruction and the intended relationships is a plausible explanation. Scrum's reference was itself RNM-derived; the contributor-count example shows how fitted smoothing can weaken a constraint. Restricted anchor and search diagnostics support this explanation for the Scrum–Chrome contrast, without proving its cause or explaining every BN. Reference construction, influence assumptions, elicited judgments, and optimization can all contribute. Blomaard et al.'s comparison optimizes against known CPTs [2], while our experimental fits use only LLM judgments. Those designs address different limits of partial elicitation. The seven-BN results caution against inferring numerical fidelity from ordinal eligibility alone.

RQ2: Elicitation support. Structured prompting modestly improves node-weighted direct elicitation, but adjusted significance does not persist under equal-family weighting. It supplies no demonstrated benefit for RNM. Nafar et al. distinguish changes in probability-query granularity from prompt modifications [18]; their within-row elicitation gains do not establish an advantage for omitting rows and reconstructing tables. Our qualitative RAG adds no demonstrated improvement beyond shared definitions. Clinical modeling support [6] and CPT reconstruction from statistical summaries [12] involve different evidence and endpoints. Local measurements may constrain probabilities in ways that our qualitative cards cannot.

RQ3: Accuracy and effort. The human burden–fidelity tradeoff motivates partial elicitation [1], but fewer requested judgments do not translate proportionally into LLM token savings. Shared inputs dominate this implementation's workload. The observed 12.1% output reduction accompanies only 0.61% fewer total tokens and higher mean CPT error. This is evidence about the tested request design, not a claim that all RNM implementations have the same cost profile.

5.2 Implications for SE researchers and practitioners

Researchers should evaluate the complete constructed artifact, retain every eligible case, and show variation behind aggregate scores. Stating the weighting rule makes clear whether a result describes the sampled nodes or equally weighted application families. Anchor comparisons can diagnose fitting changes, but should complement final-table evaluation. Work combining neural judgments and structured inference, such as BIRD [9], also motivates evaluating downstream predictions and decisions instead of assuming that inspectability guarantees usefulness.

For practitioners, direct elicitation is the stronger average numerical baseline in this sample; RNM remains a candidate when its assumptions fit the intended relationships or output burden is the priority. Neither method is validated for deployment by CPT agreement alone. Check meanings, influence directions, constraints, and probabilities against organizational measurements or experts. A small RNM fitting residual shows agreement with the elicited judgments, not with the organization. Reusable BN components may support initialization, but tailoring and held-out organizational validation remain necessary before decision use.

6 Threats to Validity

Construct validity. TV measures contextual CPT agreement, not observed calibration, expert replacement, explanation quality, or decision utility. Equal-row scoring includes potentially rare configurations. Scrum's reconstruction lacks an independently certified AgenaRisk export, and its RNM origin favors the tested representation. The seven-BN breakdown and restricted fitting diagnostics expose this distinction. SESSI represents an original elicitation stage. Adapted variable meanings and uncertain parent directions may affect agreement despite correct indexing.

Internal validity. Fresh sessions, shared definitions, reference isolation, and blinded fitting limit direct access leakage. Public benchmark memorization remains possible. The logs identify the requested model configuration, but cannot independently verify the exact serving-model version. Prompt and fitting development used the Scrum setting, which may favor that reference. Offline variants cannot exhaust all explanations. Reference-informed selection is optimistic in-sample diagnosis; bounded search may miss better fits. Schema validation cannot establish semantic or evidence fidelity, and exhaustive human review was not performed. The elicitation instructions and evidence cards were tested as bundles, so the contribution of each individual instruction or passage is unknown.

Conclusion validity. Five blocks give little evidence about the distribution or stationarity of differences. The paired-t and exact-sign results must be considered together; neither treats shared nodes or rows as independent replications. The SE restriction, completetable primary endpoint, weighting choices, and statistical plan were selected after inspecting earlier results; inference is therefore exploratory. Separate Holm families do not provide study-wide confirmatory error control. No prespecified acceptable-error margin supports claims of equivalence or practical sufficiency. Repeated execution estimates here are conditional on these fixed references and settings.

External validity. The 61-node sample spans seven BNs but only five publication families; Scrum and Chrome supply 52 nodes. Equal-family sensitivity changes their influence without making the sample representative. Some nodes offer no reduction in elicited rows, and tables have at most 27 rows. Only one requested model and reasoning effort were evaluated. Organizational distributions, larger CPTs, other model families, and evidence containing local numerical measurements may yield different tradeoffs.

7 Conclusion

We compared direct LLM elicitation with LLM-assisted RNM on complete CPTs for 61 ordinal nodes from seven SE BNs, including anchors. Eight conditions varied method, prompting, and qualitative RAG over five replications.

Direct elicitation achieved lower mean TV: 0.2317 versus 0.2404, a 3.76% increase with RNM. Equal publication-family weighting retained the direction and increased the difference. RNM nevertheless improved 31 of 61 nodes and the mean for Scrum and SESSI; larger deteriorations elsewhere outweighed those gains. Reconstruction therefore cannot be recommended from ordinal eligibility or reduced input size alone. Both overall weighting contrasts passed Holm-adjusted paired t-tests, but not exact sign tests with five replications, limiting the strength of the statistical evidence.

Structured prompting provided a modest benefit for node-weighted direct elicitation, without consistent statistical support across weighting rules. Qualitative RAG demonstrated no benefit. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, yet only 0.61% fewer total tokens. Its savings in returned information thus accompanied higher mean complete-table error, rather than establishing a superior accuracy–effort tradeoff.

Researchers should report complete-pipeline comparisons, weighting sensitivity, and network variation. For practitioners, LLM-generated CPTs remain candidate initializations requiring validation of probabilities and constraints in the intended organization. Future work should test local quantitative evidence, additional model configurations, larger CPTs, and prediction or decision quality on held-out projects. This would connect reference agreement to explainable SE decision support.

Replication Materials

The replication package contains the elicitation materials, reference CPTs, saved responses, fitted tables, evaluation scripts, token counts, and supplementary analyses. Reference materials are separated from collection inputs. The repository is private for author review.

Use of Generative AI

LLM judgments are experimental data (Section 3). Generative AI assisted drafting and script preparation/checking; numerical claims trace to saved responses and deterministic analyses.

References

[1] Martine J. Barons, Steven Mascaro, and Anca M. Hanea. 2022. Balancing the Elicitation Burden and the Richness of Expert Input When Quantifying Discrete Bayesian Networks. *Risk Analysis* 42, 6 (2022), 1196–1234. doi:10.1111/risa.13772

[2] Bodille P. M. Blomaard, Gabriela F. Nane, and Anca M. Hanea. 2025. Approaches for Reducing Expert Burden in Bayesian Network Parameterization. *Entropy* 27, 6, Article 579 (2025), 27 pages. doi:10.3390/e27060579

[3] Sicong Cao, Xiaobing Sun, Ratnadira Widyasari, David Lo, Xiaoxue Wu, Lili Bo, Jiale Zhang, Bin Li, Wei Liu, Di Wu, and Yixin Chen. 2025. A Systematic Literature Review on Explainability for ML/DL-based Software Engineering. *Comput. Surveys* 58, 4, Article 95 (2025), 34 pages. doi:10.1145/3763230

[4] Alexander Capstick, Rahul Krishnan, and Payam Barnaghi. 2025. AutoElicit: Using Large Language Models for Expert Prior Elicitation in Predictive Modelling. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 6746–6777. https://proceedings.mlr.press/v267/capstick25a.html

[5] Kwai-Sang Chin, Da-Wei Tang, Jian-Bo Yang, Shui Yee Wong, and Hongwei Wang. 2009. Assessing new product development project risk by Bayesian network with a systematic probability generation methodology. *Expert Systems with Applications* 36, 6 (2009), 9879–9890. doi:10.1016/j.eswa.2009.02.019

[6] Mario A. Cypko, Muhammad Agus Salim, Aditya Kumar, Leonard Berliner, Andreas Dietz, Matthaeus Stoehr, and Oliver Amft. 2026. Large language models with retrieval-augmented generation enhance expert modelling of Bayesian network for clinical decision support. *International Journal of Computer Assisted Radiology and Surgery* 21, 2 (2026), 211–222. doi:10.1007/s11548-025-03524-9

[7] Artur d'Avila Garcez and Luis C. Lamb. 2023. Neurosymbolic AI: the 3rd wave. *Artificial Intelligence Review* 56, 11 (2023), 12387–12406. doi:10.1007/s10462-023-10448-w

[8] Santiago del Rey, Silverio Martínez-Fernández, and Antonio Salmerón. 2023. Bayesian Network analysis of software logs for data-driven software maintenance. *IET Software* 17, 3 (2023), 268–286. doi:10.1049/sfw2.12121

[9] Yu Feng, Ben Zhou, Weidong Lin, and Dan Roth. 2025. BIRD: A Trustworthy Bayesian Inference Framework for Large Language Models. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=fAAaT826Vv

[10] Norman Fenton, Martin Neil, William Marsh, Peter Hearty, Łukasz Radliński, and Paul Krause. 2008. On the effectiveness of early life cycle defect prediction with Bayesian Nets. *Empirical Software Engineering* 13 (2008), 499–537. doi:10.1007/s10664-008-9072-x

[11] Norman E. Fenton, Martin Neil, and Jose Galan Caballero. 2007. Using Ranked Nodes to Model Qualitative Judgments in Bayesian Networks. *IEEE Transactions on Knowledge and Data Engineering* 19, 10 (2007), 1420–1432. doi:10.1109/TKDE.2007.1073

[12] Jonas Gottal and Florian Matthes. 2026. Verifiable Parameterization of Bayesian Networks from Scientific Literature: Unlocking Unstructured Empirical Evidence. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 3400–3414. doi:10.18653/v1/2026.findings-acl.167

[13] Jan-Phillip Herrmann, Sven Tackenberg, Elio Padoano, Jörg Hartlief, Jens Rautenstengel, Christine Loeser, and Jörg Böhme. 2022. An ERP Data Quality Assessment Framework for the Implementation of an APS system using Bayesian Networks. *Procedia Computer Science* 200 (2022), 194–204. doi:10.1016/j.procs.2022.01.218

[14] Andrey Krutauz, Tapajit Dey, Peter C. Rigby, and Audris Mockus. 2020. Do code review measures explain the incidence of post-release defects? Case study replications and Bayesian networks. *Empirical Software Engineering* 25 (2020), 3323–3356. doi:10.1007/s10664-020-09837-4

[15] Pekka Laitila and Kai Virtanen. 2020. On Theoretical Principle and Practical Applicability of Ranked Nodes Method for Constructing Conditional Probability Tables of Bayesian Networks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 50, 5 (2020), 1943–1955. doi:10.1109/TSMC.2018.2792058

[16] Martí Manzano, Claudia Ayala, Cristina Gómez, Antonin Abherve, Xavier Franch, and Emilia Mendes. 2021. A Method to Estimate Software Strategic Indicators in Software Development: An Industrial Application. *Information and Software Technology* 129 (2021), 106433. doi:10.1016/j.infsof.2020.106433

[17] Emilia Mendes. 2012. Using knowledge elicitation to improve Web effort estimation: Lessons from six industrial case studies. In *Proceedings of the 34th International Conference on Software Engineering*. IEEE, 1112–1121. doi:10.1109/ICSE.2012.6227108

[18] Aliakbar Nafar, Kristen Brent Venable, Zijun Cui, and Parisa Kordjamshidi. 2026. Extracting Probabilistic Knowledge from Large Language Models for Bayesian Network Parameterization. *Transactions on Machine Learning Research* (May 2026). https://openreview.net/forum?id=Fy3Byg3CVo

[19] Mirko Perkusich, Kyller Gorgonio, Hyggo Almeida, and Angelo Perkusich. 2017. A Framework to Build Bayesian Networks to Assess Scrum-based Software Development Methods. In *Proceedings of the 29th International Conference on Software Engineering and Knowledge Engineering*. 7 pages. doi:10.18293/SEKE2017-139

[20] Mirko Perkusich, João Nunes, Emilia Mendes, Emanuel Dantas, Ademar Sousa, Danyllo Albuquerque, Kyller C. Gorgônio, and Angelo Perkusich. 2026. Semi-Automatic Quantification of Bayesian Networks for Software Decision Support: Comparing WSA and RNM in a Software R&D Organization. *arXiv preprint arXiv:2609.31275* (2026), 37 pages. https://arxiv.org/abs/2609.31275

[21] Mirko Perkusich, Gustavo Soares, Hyggo Almeida, and Angelo Perkusich. 2015. A procedure to detect problems of processes in software development projects using Bayesian networks. *Expert Systems with Applications* 42, 1 (2015), 437–450. doi:10.1016/j.eswa.2014.08.015

[22] Kumar Rahul and Shovan Chowdhury. 2026. Human–AI Construction of Bayesian Networks for Operational Decision Support: A Virtual Survey Approach. *arXiv preprint arXiv:2607.14141* (2026). doi:10.48550/arXiv.2607.14141

[23] Thiago Rique, Mirko Perkusich, Emanuel Dantas, Danyllo Albuquerque, Kyller Gorgônio, Hyggo Almeida, and Angelo Perkusich. 2023. On Adopting Software Analytics for Managerial Decision-Making: A Practitioner's Perspective. *IEEE Access* 11 (2023), 73145–73163. doi:10.1109/ACCESS.2023.3294823

[24] Thiago Rique, Mirko Perkusich, Kyller Gorgônio, Hyggo Almeida, and Angelo Perkusich. 2025. Constructing the graphical structure of expert-based Bayesian networks in the context of software engineering: A systematic mapping study. *Information and Software Technology* 177 (2025), 107586. doi:10.1016/j.infsof.2024.107586